

FAIRGAME: A Framework for AI Agent Bias Recognition Using Game Theory

- Problem: Multi-agent LLM systems are hard to interpret and predict, and hidden biases can skew strategic outcomes (e.g. negotiation, pricing, dispute resolution).
- Gap: Game theory can model strategic interaction, but there's no standardized, reproducible tool for testing LLM agents in games.
- Contribution: An open-source framework to simulate games, vary agent traits, and compare results against game-theoretic predictions.

How the Framework Works

- **Two inputs:**
 - A JSON config file (payoff matrix, rounds, personalities, languages, LLM choice, whether agents know the round count)
 - A prompt template (a text description of the game, translatable into any language)
- **Pipeline:** Config and template are validated, agents are instantiated on the chosen LLM, games are created and run round by round, and the history and payoffs are saved.
- **Key knobs:** LLM, language, personality (cooperative or selfish), knowledge of the opponent's personality, and knowledge of the number of rounds

Experimental Setup

- **Games:**
 - Prisoner's Dilemma in 3 variants (conventional, harsh, mild), which change the dilemma strength
 - Battle of the Sexes (coordination game)
- **LLMs:** Llama 3.1 405B, Mistral Large, GPT-4(o), Claude 3.5 Sonnet
- **Languages:** English, French, Arabic, Vietnamese, Mandarin (machine-translated, then checked by native speakers)
- **Conditions:** 10 rounds per game, rounds known or unknown, personalities CC / CS / SS, and agents unaware of the opponent's personality in these tests
- **Scale:** 18 games per LLM, each repeated 10 times resulting in approx. 72,000 decisions in total

Key Findings

- **Prisoner's Dilemma:**

- Agents mostly defect, in line with the Nash equilibrium, but behavior varies by language and personality, so it isn't driven by payoffs alone.
- Knowing the number of rounds increases cooperation in CC and CS settings, but not in SS.
- Harsher payoffs lead to more defection and milder payoffs to more cooperation.

- **Battle of the Sexes:**

- Agents generally coordinate, but selfish personalities sharply reduce cooperation.
- Claude 3.5 approaches near-perfect coordination, while Llama 3.1 405B stays largely uncoordinated.

- **Do LLMs just know the games?**

- Assigning Hitler vs. Gandhi identities produced betrayal vs. cooperation, which suggests the agents use prior knowledge beyond the payoff matrix.

Scoring System, Conclusions & Future Work

- 4 normalized metrics shown as plots:
 - Internal Variability (consistency across repeated runs)
 - Cross-Language Inconsistency
 - Sensitivity to Payoff
 - Variability Over Rounds
- Reported patterns: In the Prisoner's Dilemma, Mistral Large has the highest variability, GPT-4o the highest payoff sensitivity, and Llama 3.1 405B the most stable behavior.
- Takeaway: LLM agents deviate from game-theoretic predictions in ways that depend on language, persona, and context, so reproducible simulation pipelines are needed.
- Future work:
 - Bias dimensions beyond language and personality (nationality, gender, etc.)
 - Agent communication & More than 2 agents
 - Incomplete-information and sequential games